

PROJECT FILE

Auto Ticket Classification using Flow

Sample academic project documentation • System design and workflow

SAMPLE PROJECT DOCUMENTATION

Prepared as an academic project reference

Abstract

Customer-support teams receive many tickets through email, forms, and service portals. Manually reading and assigning each ticket can delay responses and create inconsistent routing. This project demonstrates an automated text-classification workflow that reads ticket descriptions and predicts a support category. Ticket text is cleaned, converted into TF-IDF features, and passed to a supervised machine-learning classifier. The predicted category can then be used to route the ticket to the relevant team. Human review and updated training examples are included as a feedback loop.

1. Introduction

Auto Ticket Classification using Flow is a small decision-support application for organizing incoming service requests. It focuses on categorizing ticket text rather than automatically resolving the customer's issue. Example categories are Billing, Technical Support, Account Access, and Delivery.

2. Problem Statement

When ticket volume increases, manual classification consumes staff time and may lead to tickets being assigned to the wrong team. A repeatable automated workflow can provide an initial category and help staff prioritize routing while keeping a human review option.

3. Objectives

- Accept a ticket description as text.
- Prepare text consistently for machine-learning processing.
- Predict one category from the categories represented in labelled training data.
- Show the predicted category and a model confidence score.
- Support review of incorrect predictions and periodic model updates.

4. Scope and Limitations

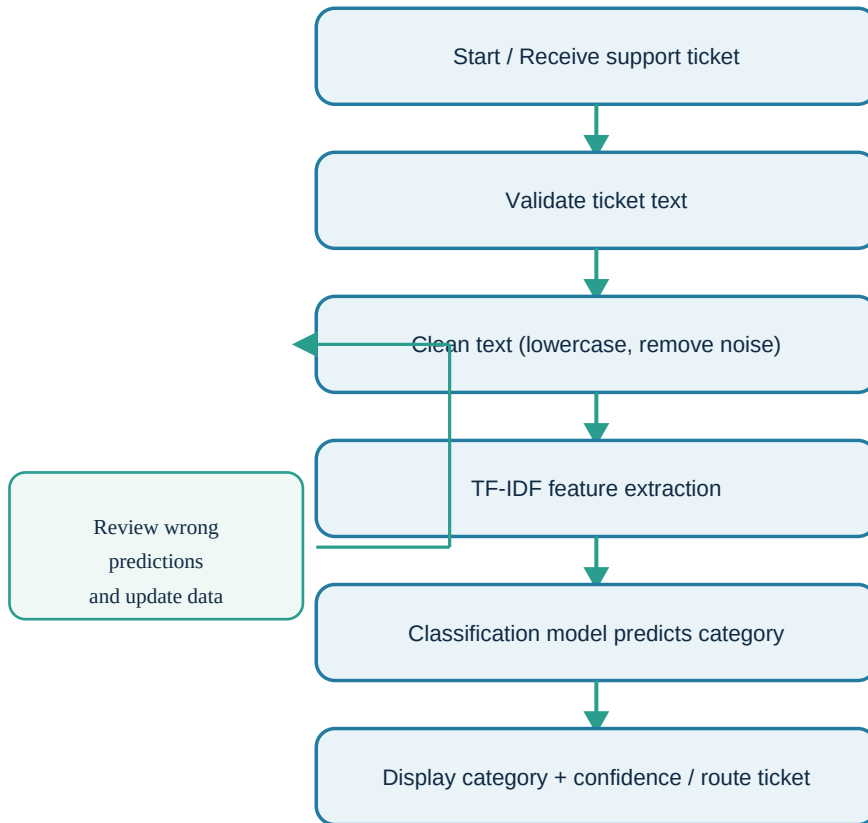
The prototype is intended for academic demonstration and local testing. It does not include authentication, a production ticket database, email integration, service-level prioritization, or automatic responses. Prediction quality depends on the amount, balance, and correctness of labelled training data. A staff member should verify uncertain or sensitive tickets.

5. System Requirements

Type	Requirement
Hardware	Computer with 4 GB RAM or more; internet only for installing packages.
Software	Python 3; pandas; scikit-learn; joblib; a code editor or terminal.
Input	CSV containing ticket text and its labelled category.
Output	Predicted ticket category and classifier confidence score.

6. Proposed System Flow

The workflow begins when a support ticket is received. The text is checked, normalized, transformed into numerical TF-IDF features, and classified using a model trained on labelled examples. The result is displayed to the support agent for routing. Incorrect predictions can be reviewed and added to the training dataset for a later model update.



7. Modules

Module	Function
Ticket Input	Receives a ticket description from the user or service portal.
Text Preprocessing	Handles missing text and standardizes input before vectorization.
Feature Extraction	TF-IDF represents words and word pairs as numeric features.
Model Training	Logistic Regression learns patterns from labelled ticket examples.
Prediction and Routing	Predicts a category and displays a confidence score for staff review.
Feedback and Retraining	Staff-corrected labels can be added to the dataset for future training.

8. Algorithm

1. Load labelled ticket examples from the CSV file. 2. Remove rows without text or category labels. 3. Split the data into training and test sets when there are enough examples per category. 4. Convert ticket text into TF-IDF vectors using unigrams and bigrams. 5. Train a Logistic Regression classifier. 6. Evaluate on held-out examples when a valid split is possible. 7. Save the trained pipeline and use it to predict the category of a new ticket.

9. Testing Plan

Test case	Input / action	Expected result
Billing ticket	"I was charged twice."	Category predicted from trained labels; typically Billing if examples support it.
Account access	"I forgot my password."	Category predicted from trained labels; typically Account Access if examples support it.
Blank input	Press Enter without text.	Show message that no ticket text was entered.
Missing CSV	Run without tickets.csv.	File-not-found error identifies missing training file.

10. Expected Result

For a ticket whose wording resembles examples in the training dataset, the program prints a predicted category and confidence score. Results may be inaccurate for unfamiliar wording, overlapping categories, or limited training data; staff review remains part of the proposed workflow.

11. Future Enhancements

- Connect the classifier to a web-based ticket submission form.
- Store tickets, predictions, corrections, and timestamps in a database.
- Add a review queue for low-confidence predictions.
- Track category-wise precision, recall, and F1-score on a larger test set.
- Retrain the model periodically using reviewed ticket examples.

12. Conclusion

The project presents a clear flow for classifying support tickets with machine learning. TF-IDF and Logistic Regression provide a straightforward baseline that can be trained and tested locally. With representative labelled data, human feedback, and ongoing evaluation, the prototype can be extended into a more complete ticket-management system.

References

1. scikit-learn documentation: Text feature extraction and Logistic Regression — <https://scikit-learn.org/stable/> 2. pandas documentation — <https://pandas.pydata.org/docs/> 3. joblib documentation — <https://joblib.readthedocs.io/>

Customization note: Replace the sample categories, dataset, and test cases with the requirements and data approved for your own project.